(8)

project (Reverse car parking assistance system USING CAN protocol)

=) AIM:- when ever we are reverse parking the car we can't see the objects clearly in rear view mirror in Rainy seasons due to fog bcz of this we cars are getting damaged. ~~accurately~~ for that if we know the exact distance of object/wall is there by ~~beer~~ showing the variations in a beep sounds of buzzer. my project will generate this beep sounds by calculations ~~How~~ the ~~much~~ distance of the object (i.e) by using HC SR04 Ultrasonic sensor & ~~for~~ ~~Node~~ CAN protocol for comm'n b/w Nodes.

Requirements:- Good understanding of CAN protocol, Interrupts, Behaviour of the SR04 Ultrasonic sensor, UART protocol, Keil software ~~is a~~ (for writing code & debugging the code)

=) ~~ARM~~ Two ALPC2129 ARM controllers, LED's.

=) what I had done:-

=) To ~~acheive~~ accomplish ~~this task~~

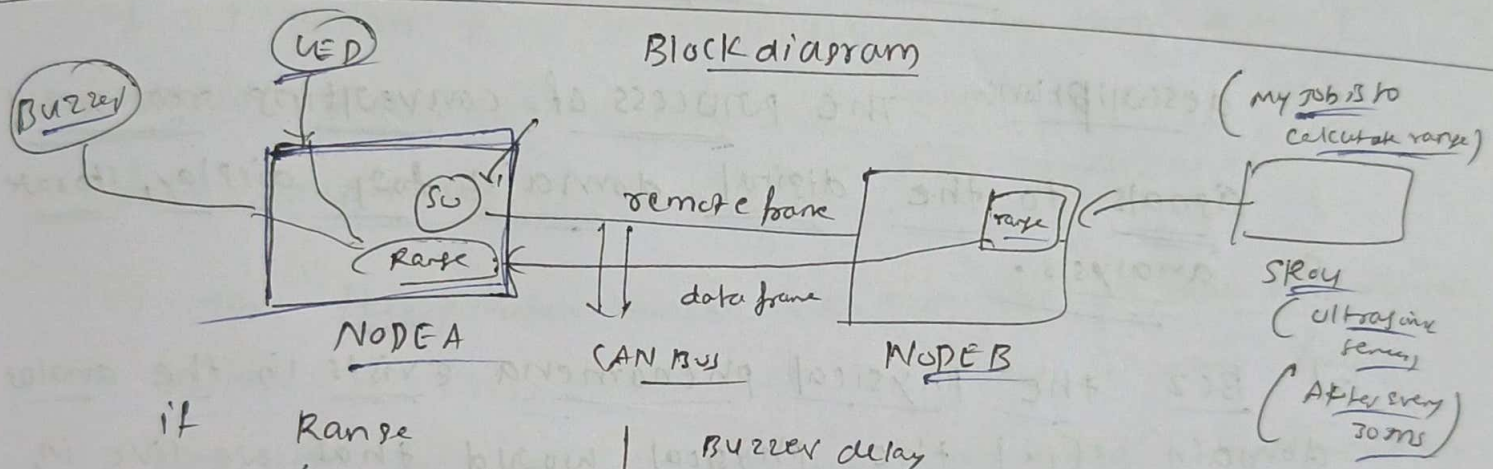
This project is implemented by two nodes, NODE-A acts as a dashboard control in which a switch is provided with External Interrupt for Enabling & disabling the reverse parking option. whenever we are Enabling the reverse parking option (i.e If we press that switch odd number of times) then the NODE A sends remote frame to the NODE B via CAN BUS, upon receiving the remote frame by the NODE B, the ultrasonic sensor is attached to the NODE B it gets activated & analysizes the distance b/w the vehicle & the wall & after the successful analysis the result of analysis is send to NODE A as data frame by NODE B, After the reception of dataframe NODE-A analysize the data (distance) & the buzzer & LED's attached on the NODE-A.

Based on the Range the b, I beeped the buzzer & blinked the LED like

if Range is b/w 200cm to 300cm | beep the buzzer after every 1 sec

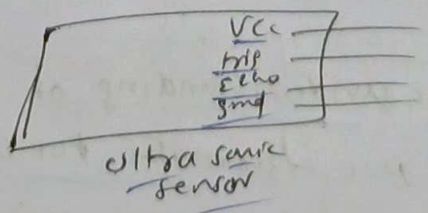
if Range is decreasing the beep sound of the buzzer will increase by observing the variance in beep sound the driver can predict the distance b/w car & the wall & he will park the car carefully without any damage by this Assistance system.

Block diagram



if Range	Buzzer delay
400cm - 300cm	1 sec
300cm - 200cm	180 ms
200cm - 100 cm	50 ms
< 50cm	Continuously beep the buzzer

=) Working of ~~SR-HC04~~ SR04:-



=) one trigger pulse is generated by us (i.e by giving 10us delay) this pulse will transmitt from sensor & when

the transmitted signal hits the wall it will sends echo signal the time ~~ia~~ to reach the echo signal to the sensor is calculated & distance is calculated by using formula

⇒ distance = speed × time.

= (speed of sound)

$d = 343 m/s \times t$

⇒ we will ~~get distance from~~ this this sensor after calculating distance (Range) it will give NODE B.

· DATA AQUISITION:-

⇒ description:- The process of converting real-world signals to the digital domain for display, storage & analysis.

⇒ BCZ the physical phenomena exists in the analog domain (i.e) the physical world that we live in, they must be first measured there & then converted to digital domain.

TOOLS:- External ADC (MCP3204), Keil software, LPC2129 version ARM controller, ~~Basic~~ LCD, LDR sensor

Basic understanding:- The basic understanding of ADC, ~~the~~ Analog to digital converter, formulas for conversion, commands in LCD to operate/display.

~~What~~ ~~Ø~~ Explanation:-

=) ~~Ø Initial~~

Initially I ~~took one sensor~~ Iam selecting one External adc (mcp3204) ~~to~~ for that Iam proving analog value & ~~Ø~~ & It is having 4 channels with 12bit resolution & Internally this ADC by using SFR's ~~converting this Real time signal into digital~~ digitize this signal & the digitized signal is given to the LCD of 16x2 resolution & It will display that value on the LCD in digital

=) ~~I used 3 sensors~~ I tried to display three values digitized values from three different sensors on LCD (i.e) from temperature sensor, LDR sensor, potentiometer.

=) The temperature sensor will give the ~~Ø~~ value of digital voltage & this value will be useful in many further projects for ~~Ø~~ example for dashboard Implementation if we need to display the temperature ~~val~~ of the car on a dashboard (Instrumental cluster) ~~we~~ definitely the data aquisition is needed.

=) & The LDR sensor (photo) it will detect the amount of light by using the intensity of light, & It is having Internally one ~~Resistor that Resistor~~ two conductors seperated by an Insulator. when light hits the LDR the insulator becomes more conductive, which decreases the Resistance & allows more current to pass through.

=) like this some sensor values can be digitized by using data aquisition.

Applications :-

1. In Environmental Monitoring.
2. Research & development.
3. Transportation.
4. [illegible]

Introduction :-

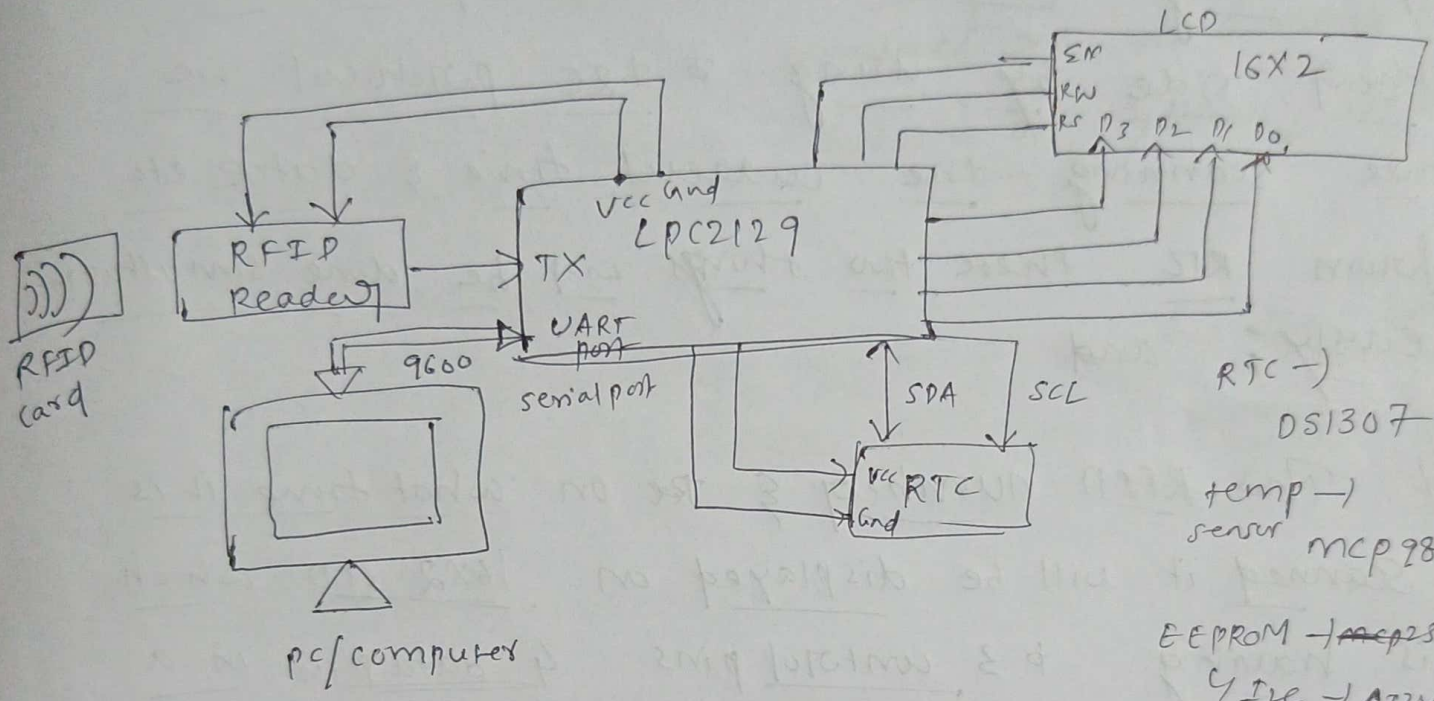
Good ~~afternoon~~ morning sir

It's my pleasure to introduce [illegible] to

digital electronics:-

⇒ RFID Attendence system:-

(Employee - Attendence system using RFID):-



RTC → DS1307

temp → sensor MCP9808

EEPROM → ~~MCP25~~ I2C → AT24C01

⇒ ~~Here we~~ AIM:- To monitor time, date & card number of employee & store it into database.

⇒ TOOLS:- Keil software, LPC2129 µc'er, RTC (DS1307)

H/w requirements: 16X2 LCD, RFID card, RFID Reader, pc.

Basic ~~knowld~~

Software requirements:- Basic knowledge of c, embedded c, I2C protocol, Uart protocol, RTC & RFID Reader functionality

=) ~~An employee by~~
first we need to scan
~~while scanning~~ a 12bit RFID card which is having a 12bit number using RFID Reader this RFID Reader is connected to LCP2129 uCtr by using UART ~~transmission~~ protocol. & on the other side by using ~~a~~ I2C protocol we are scanning the current time & date etc from RTC these two things will be done simultaneously; ~~and~~

=) This RFID number & ~~Re~~ on what time it is scanned it will be displayed on 16X2 LCD which is having ~~a~~ 3 control pins 4 data pins in a 4 bit enabled mode. & we ~~need~~ to as per our goal we need to store this information in ~~the~~ a database which is present in our PC.

=) ~~one manu~~ we will create one manual file & we will try to compare ~~the~~ two strings ~~which is one is to~~ the one which is ~~presented~~ present in a manual file & other string which is formed as a mixture of RTC o/p & RFID Reader o/p & both are matching in ~~Even~~ odd number of times then ~~he is it is this~~ he is the member of this company/institute & for odd times it is

his In time, if it is Even number of times then it is ~~out~~ out time. this with Everything will work on single day basis & all this information will stored into log file.

(logfile:- A logfile is a file that contains a record of Events, transactions (or) activities that have occured within a system, app (or) s/w.

=> log files are used to track & monitor the behavior of a system, diagnose problems & analyse performance.)

DASHBOARD IMPlementation using CAN protocol:-

=> AIM:- The main purpose of this project is to show the essential sensor information on the dashboard, like temp, speed & lights

=> with this Information we can control the temperature Inside the car & also Engine temp.

=> The Main advantage ~~of~~ is without Indication of ~~Rights~~ light the driver does not know is

SDLC (software development life cycle)

⇒ some companies will have 5 stages & some companies will have 8 stages but many companies will follow 7 stages.

1. Requirements gathering ✓
2. Analysis & planning. ✓
3. Design & Architecture ✓
4. Development. ✓
5. Testing. ✓
6. Integration & Deployement. ✓
7. Maintainence. ✓

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① Requirements & Gathering:-

⇒ suppose two companies.

Abcd) client company.

1st way

Sales & Marketing ~~team~~ team will ask client that ~~we will~~ tell us if you need a new software we can give you a best software.

2nd way

Clients will approach abcd to ask to design a new software via Email.

then abcd company will follow 7 steps (i.e) known as "sdlc"

1st stage is

① Requirements gathering:-

=) In this step the "bussiness analysts" will conducts a meeting with client management team & discusses like

① what kind of software's is needed.
② what are the functionalities required. & features required.

on the basis of this they will prepare one document called "BRS" (Business requirement specification).

Note:-
=) This BRS will send to client company & to the testing team of same own company also. then client company will reply reconfirm this. (for the sake of clarity)

② Analysis & planning:-

=) The business Analysists will conduct one meeting in their own company, In this meeting N/w administrators, Devops team, testing team all team leads/ managers will attend

in this meeting, the business analysis will explain about the project

=) & And every team ~~asked~~ will ask their questions & prepare their respective document called "FRS/SRS" (In this how many developers are there & how many ~~the~~ testers, time stamps. are there in this project, how many servers we need etc will present in this "FRS/SRS" Billing also.

FRS → functional requirement specification

SRS → system requirement specification

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③ Design & prototyping :-

=) In this stage a blue print will be prepared by "business Analysis" & receives conformation emails from every team.

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④ Development :- ⑤ testing

=) The development team will start coding in this stage. & ~~testi~~ testers will test this code ~~at~~ simultaneously while developers developing ~~which~~ it ~~ne~~ may be manual testing / automation testing.

=) Both development & testing teams will work simultaneously.

⑥ Integration & deployement :-

=) In this stage our ~~devops~~ devops team, data admin s~~d~~ team, N/L admin team, Aws Cloud archetect will involve in this stage.

=) They will convert that developers code into live ~~Ent~~ Environment code.

=) This ~~Envirg~~ live Environment code move into client servers, This shifting (shift) called as " Deployment "

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⑦ maintainence :-

=) To This will provide security, ~~to the~~ live ~~servers soft~~ bc2 the software is live, is there any ~~Hacker~~ Hacking attack is there, is the database is updating properly (or) not. etc

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